Vaibhav Saini

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SUMMARY

Early career AI and Full Stack Software Engineer with hands on experience building production RAG systems, LLM cost routers, and async API backends from scratch. I care about latency, failure modes, and whether Al output can actually be trusted. I move fast, take full ownership, and have independently designed and deployed AI systems end to end. But I never stop thinking about whether something feels intuitive to the person on the other end. On OPT, looking for a full-time role.

EXPERIENCE

Software Engineer

Badoliasoft **June 2023 - July 2024**

- Implemented JWT auth with HMACSHA512 hashing, 3 tier RBAC, and token blacklisting, securing a fintech platform for 100+ users with zero unauthorized access incidents post-deployment.
- Built real time notification pipeline using SignalR with RabbitMQ fanout exchange fallback; integrated Azure Blob storage with CDN, reducing average file delivery latency by 40%.
- Configured IP-based rate limiting, global exception middleware with structured logging, and Redis distributed caching, reducing database load by 35% and improving API reliability under peak traffic.
- Contributed to a full-stack Angular + .NET 8 platform (over 200 C# files, 10 REST controllers) built with TypeScript and JavaScript; spanning user management, messaging, and project workflows.
- Operated as a core contributor on a small team building a greenfield fintech platform from scratch, owning auth, notifications, and caching systems across the full delivery cycle.

PROJECT

SecRAG Pro

github.com/vsaini06/SecRAG-Pro • September 2025 - April 2026

- Merged two independently built systems, a hybrid RAG pipeline and an LLM cost router, into one unified production backend, eliminating redundant inference calls and centralizing routing logic across GPT-4o, GPT-4o Mini, Claude Sonnet, Claude Haiku, and local Llama 3.2.
- Achieved a 62.3% cost reduction versus all-GPT-4o routing across 513 test queries with zero failures, routing 41% of queries to free local inference via Ollama with no measurable quality degradation.
- Built hybrid BM25 and dense vector retrieval fused via Reciprocal Rank Fusion with a cross-encoder reranker, cutting the top 20 candidates to the top 5 and achieving a 4.15/5.0 average answer quality on a 48-question golden evaluation set.
- Validated answer trustworthiness through LLM-as-judge verification checking every claim against source chunks, achieving 99.6% citation accuracy for high-stakes document queries.
- Extended the system into LegalAdaptRAG, an intent-aware adaptive retrieval pipeline for legal documents with 5-class intent classification, per-intent retrieval strategy selection, and confidence scoring, targeted at users who cannot afford legal representation.

EDUCATION

Master of Science in Information Technology (Security)

Arizona State University • Tempe, AZ • 2026 • 3.97 / 4.0

Bachelor of Engineering in Computer Science (Data Science)

MMDU • Ambala, Haryana, India • 2023 • 7.62 / 10.0

CERTIFICATIONS

Machine Learning Specialization

DeepLearning.AI • 2026

Generative Al and LLMs

DeepLearning.AI • 2026

AWS Academy Cloud Architect and Solutions Engineer

Amazon Web Services • 2025

AWS Certified Cloud Practitioner

Amazon Web Services • 2025

SKILLS

Technical Skills: React, Angular, Python, Java, TypeScript, SQL, ASP.NET Core, C#, Docker, GitHub, Git, Machine Learning, Algorithm Design, LLMs, Node.js, Rest APIs